On the relation between Uniform Information Density and speakers' traits

According to the Uniform Information Density (UID) hypothesis [1-2], speakers prefer utterances in which information is spread evenly across the linguistic signal. Despite abundant empirical evidence for UID across multiple levels of linguistic representations [1, 3-4], the range of reasons for UID effects arise are less well-understood. Here we explore the hypothesis that a communicative need to facilitate listener comprehension contributes towards UID effects. To assess this possibility, we test whether and how information distributions are influenced by individual differences in speakers' personality traits. Specifically, we hypothesize that if UID effects stem at least partly from speakers taking listeners' needs into account, then speakers with personality traits reflecting social and communicative ability should show stronger UID effects.

Methods Our analysis uses CANDOR [5], a corpus of 1,656 dyadic Zoom conversations from 1,456 participants, accompanied by extensive survey data on personality traits and other factors.

To tap into **speakers' personality traits**, we use several socio-pragmatic measures: the communication and social skills subscales of the Autism-Spectrum Quotient (AQ) [6], the Reading the Mind in the Eyes Test (RMET, [7]; assessing Theory of Mind and emotion inference from facial cues), the Perspective-Taking (PT) and Empathic Concern (EC) subscales of the Interpersonal Reactivity Index (IRI, [8]), and the Big 5 Inventory (BFI, [9]), which measures five personality dimensions: openness, conscientiousness, extraversion, agreeableness, and neuroticism. Distributions of the raw (non-transformed trait) scores are shown in Fig. 1.

We also compute a measure of **information density uniformity for each speaker.** Following [10, 11], a word's information density is the negative log probability of the word given its preceding context (Eq. 1), estimated using GPT-2 Small [12] with context spanning all prior words in the current and previous turns. Information uniformity is quantified as the mean squared deviation of each word's information content from the mean across a text, following [13] (Eq. 2), where *lower values indicate greater uniformity*.

To assess the relation between personality traits and UID, we fitted linear mixed-effects regression models with information uniformity as the dependent variable and standardized personality trait scores as predictors. Because all participants completed the BFI but only a subset completed the others, we fit one model for the BFI and another for other measures (Eqs. 3–4).

Predictions If sensitivity to others' mental and emotional states enhances UID effects, speakers with **higher** scores on agreeableness on the BFI (reflecting greater empathy, cooperativeness, and prosocial orientation), RMET (ability to infer others' emotional states), and the PT subscale of the IRI (ability to perspective-take) should also exhibit **stronger** UID effects. We make no predictions regarding the other BFI dimensions or the EC subscale of the IRI but included them in the model for completeness and exploratory purposes.

Similarly, for the scores on the communication and social skills subscales of AQ (scored such that lower scores indicate stronger communicative and social abilities), we predict that speakers with **lower** scores should exhibit **higher** information uniformity.

Results from the two regression models are summarized in Tables 1 and 2. As predicted, BFI Agreeableness exhibited a significant positive effect on information uniformity, indicating that individuals higher in Agreeableness produced more smoothly distributed information. RMET scores also showed a significant positive association with information uniformity, consistent with our predictions. The other measures show no significant effects. In particular, we found no evidence that social and communication skills (as measured by the AQ) modulate UID. Further, despite the IRI PT subscale measuring inferential perspective-taking, it showed no effects. A key distinction is that RMET provides a more objective measure of how well a person can infer others' mental states (identify emotions from faces), whereas IRI-PT relies on *introspective self-report* regarding a person's ability to infer others' mental states/perspectives.

In sum, our results provide the first large-scale evidence that UID is modulated by speakers' ability to infer others' mental states. The effects from Agreeableness and RMET supports the hypothesis that UID effects stem at least partly from speakers' sensitivity to listeners' needs.

Eq. 1: $I(w_i) = -\log P(w_i|c)$

Eq. 2: $U = 1/n\ ((I(w_1)-\mu(l))^2 + (I(w_2)-\mu(l))^2 + \ldots\ (I(w_n)-\mu(l))^2)$

Eq. 3: uniformity ~ Openness + Conscientiousness + Extraversion + Agreeableness + Neuroticism + (1 | speaker) + (1 | conversation)

Eq. 4: uniformity ~ AQ Communication + AQ Social Skill + RMET + IRI PT + IRI EC + (1 | speaker) + (1 | conversation)

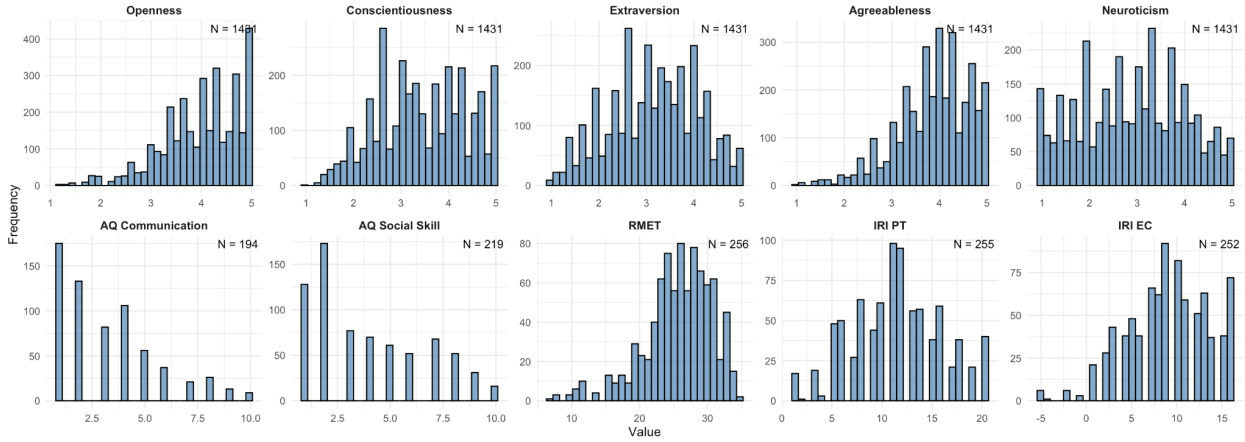

Fig. 1: Distributions of the BFI, AQ, RMET and IRI scores across participants.

Table 1: Effects of the BFI on information uniformity (GVIF values show no multicollinearity)

Predictor	Estimate	Std. Error	t value	p value
(Intercept)	0.0009	0.023	0.040	0.968
Openness	-0.013	0.023	-0.584	0.560
Conscientiousness	0.036	0.026	1.384	0.167
Extraversion	-0.011	0.024	-0.482	0.630
Agreeableness	0.061	0.024	2.593	0.010**
Neuroticism	0.033	0.027	1.238	0.216

Table 2: Effects of AQ, RMET, and IRR on information uniformity (GVIF values show no multicollinearity)

Predictor	Estimate	Std. Error	t value	p value
(Intercept)	-0.086	0.057	-1.506	0.134
AQ Communication	0.019	0.071	0.272	0.786
AQ Social Skill	0.030	0.073	0.417	0.678
RMET	0.134	0.053	2.510	0.013*
IRI PT	-0.010	0.065	-0.162	0.872
IRI EC	-0.098	0.062	-1.576	0.117

References: [1] Jaeger (2010) Cogn. Psychol. [2] Levy & Jaeger (2007) NeurIPS. [3] Aylett & Turk (2004) Lang. Speech [4] Mahowald et al. (2013) JML. [5] Reece et al. (2023) Sci. Adv. [6] Baron-Cohen et al., 2001; [7] Baron-Cohen et al., 2001; [8] Davis, 1983; [9] John et al., 1991; [10] Shannon (1948) Bell Syst. Tech. J. [11] Levy (2008) Cognition [12] Radford et al. (2019) OpenAI. [13] Meister et al. (2021) EMNLP.